Supplementary Material: Fixed-Spectrum Swaps Separate Attention Capacity from Assignment

Anonymous submission

This document supplies proofs, experimental protocols, complete result tables, secondary figures, negative replications, and reproducibility information for the main paper. The main paper is self-contained; this supplement preserves the full audit trail behind its compressed claims.

Theory Details

Established softmax grouping identity

For evidence positions S, distractors D, scores s_j, and temperature $T > 0$, let $Z_G = \sum_{j \in G} e^{s_j/T}$ and $F_G = -T \log Z_G$. Direct grouping of the softmax denominator gives

$$m_S := \sum_{j \in S} a_j = \frac{Z_S}{Z_S + Z_D} = \sigma\left(\frac{F_D - F_S}{T}\right). \quad (1)$$

Defining the group log-mean-exp score $\bar{s}_G = T \log(|G|^{-1} \sum_{j \in G} e^{s_j/T})$ yields the exact threshold relation

$$m_S \geq \alpha \Longleftrightarrow \bar{s}_S - \bar{s}_D \geq T\left(\log \frac{|D|}{|S|} + \operatorname{logit}(\alpha)\right). \quad (2)$$

These are standard Gibbs or partition-function identities (Ramsauer et al. 2021). They motivate the competition for evidence mass and position the paper. Equation 2 also prevents an overstatement: distractor count alone is insufficient because arbitrarily many negligible-score distractors need not reduce evidence mass.

Fixed-spectrum routing range

Let $a_{(1)} \geq \cdots \geq a_{(n)}$, let S contain r evidence positions, and define

$$C_r^+ = \sum_{i=1}^{r} a_{(i)}, \qquad C_r^- = \sum_{i=n-r+1}^{n} a_{(i)}, \qquad K_r(a) = C_r^+ - C_r^-. \quad (3)$$

Theorem S1 (Capacity–assignment decomposition)**.** *For every permutation π of a fixed attention-weight multiset, $C_r^- \leq m_S(\pi) \leq C_r^+$, both endpoints are attainable, and for $K_r(a) > 0$ the unique coordinate*

$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1] \quad (4)$$

gives $m_S(\pi) = C_r^- + \rho_S(\pi) K_r(a)$.

For affine margin $M(\pi) = b + \sum_j a_{\pi(j)} u_j$, sorted utilities $u_{(1)} \geq \cdots \geq u_{(n)}$ give

$$\max_{\pi} M(\pi) = b + \sum_i a_{(i)} u_{(i)}, \quad (5)$$

$$\min_{\pi} M(\pi) = b + \sum_i a_{(i)} u_{(n+1-i)}. \quad (6)$$

No statistic invariant to token permutation can determine evidence assignment or the output margin.

Proof. An r-position set receives at most the r largest weights and at least the r smallest; assigning those weights to S attains the endpoints. Normalizing the bounded mass gives ρ_S. For the output range, suppose $u_p > u_q$ but $a_{\pi(p)} < a_{\pi(q)}$. Swapping those weights changes the margin by $(a_{\pi(q)} - a_{\pi(p)})(u_p - u_q) > 0$. Repeated exchanges align weights and utilities for the maximum and reverse them for the minimum, the classical rearrangement inequality (Hardy, Littlewood, and Pólya 1952). The extremal assignments preserve the complete weight multiset and therefore every permutation-invariant statistic. □

Nonlinear downstream network

Fix an input and selected heads H at one layer. Let $x_{hj} = O_h v_{hj}$ and $z = \sum_{h \in H} \sum_j a_{hj} x_{hj}$. For target y and competitor c, let $M_c(z) = g_y(z) - g_c(z)$. A source–donor swap has

$$\delta_h = a_{hd_h} - a_{hs}, \qquad \Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \quad (7)$$

Define local utility $u_{hj}^{(c)} = \nabla M_c(z)^\top x_{hj}$.

Theorem S2 (Nonlinear routing-margin bound)**.** *If ∇M_c is L_c-Lipschitz from z to $z + \Delta z$, then*

$$\left| \Delta M_c - \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) \right| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (8)$$

Proof. The swap changes head h by exactly $\delta_h(x_{hs} - x_{hd_h})$. The fundamental theorem of calculus gives

$$\Delta M_c = \int_0^1 \nabla M_c(z + t\Delta z)^\top \Delta z \, dt. \quad (9)$$

Subtracting $\nabla M_c(z)^\top \Delta z$ and applying Lipschitzness bounds the remainder by $\int_0^1 L_c t\|\Delta z\|_2^2 dt = L_c\|\Delta z\|_2^2/2$ (Nesterov 2004). Expanding the first-order term gives Equation 8. $\square$

Corollary S3 (Multi-evidence routing)**.** *Let S contain r evidence positions and pair every evidence position missing a top-r weight with one top-r non-evidence donor. For disjoint swap pairs P_h and $\delta_{hsd} = a_{hd} - a_{hs} \geq 0$,*

$$\Delta z = \sum_{h \in H} \sum_{(s,d) \in P_h} \delta_{hsd}(x_{hs} - x_{hd}), \tag{10}$$

and Equation 8 holds with the first-order sum extended over $(s,d) \in P_h$. Thus assigning top-r capacity to the evidence improves the margin whenever the summed evidence–donor utility gain exceeds the curvature remainder.

The decomposition and corollary are algebraic for arbitrary r. Their behavioral premise is established for one and two evidence positions in the current experiments. The three-evidence interval includes zero, and the four-evidence models miss the competence gate, so efficacy beyond $r = 2$ remains open.

The condition is sufficient, not necessary. Source-max can fail when evidence values have no greater downstream utility, patched heads cancel, or curvature dominates. This is why natural source mass is not a universal circuit-selection rule.

Controlled Experiments

Tasks and training

Argmax retrieval presents key–value pairs and requests the value paired with the largest key. **Flag retrieval** presents values with exactly one marked position and requests the marked value. Both tasks are order-invariant and have one identifiable evidence position. **Reversed decimal addition** is an order-dependent contrast. **Two-evidence modular sum** marks two digits and requests their sum modulo ten, so copying either digit is insufficient.

The primary model is a four-layer pre-norm decoder transformer with width 256, eight heads, and feed-forward width 1024. We train with AdamW, batch size 512, and warmup plus cosine decay on lengths one through five. Evaluation follows a geometric ladder to $50\times$ training length. The scale replication uses eight layers and width 512. A cell is interpreted only after training-length exact match reaches 0.8. Order-invariant cells use four seeds; the order-dependent contrast uses two. The preregistration fixes the competence gate, break-point rule, and variance-collapse check before intervention outcomes.

Measured quantities and interventions

At the answer query we record evidence mass, attention entropy, every attention norm, participation ratio, attention-output variance, and target margins. The variance intervention replaces $h \leftarrow h + z$ with $h \leftarrow h + \mathrm{LN}(z)$ after attention, holding the downstream variance of z approximately constant. The concentration intervention applies length-scaled logit sharpening.

For the fixed-spectrum test, source-max swaps the source weight with the row maximum, source-min swaps it with the row minimum, and the distractor control swaps the largest and smallest non-source weights. All conditions preserve model parameters, prompts, transformed values, the complete sorted attention row, entropy, norms, maximum, and participation ratio. The control additionally preserves source mass.

Confirmatory inference. We define one family of three primary contrasts: controlled source-max minus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max minus matched control. Exact two-sided sign-flip tests operate on independent trained-model clusters for the controlled claims and calibration-seed clusters for natural QA. Holm adjustment controls family-wise error across the three tests. Hierarchical intervals and all other reported cells are descriptive and unadjusted unless stated otherwise.

Observational and variance-control results

Of the 384 aggregate controlled points, 151 (39.3%) lie at exact accuracy ceiling. Rank analyses remain separated despite that pileup: source attention has pooled Spearman 0.854 and Kendall $\tau_b = 0.713$, versus 0.445 and 0.338 for attention-output variance. On the 233 non-ceiling points, the corresponding values are $0.912/0.751$ and $0.544/0.402$. Mean within-cell Spearman is 0.947 for attention and 0.777 for variance. The variance intervention holds the post-intervention ratio near one, yet accuracy remains impaired (Table S1).

Capacity, assignment, utility, and dose

Across sixteen trained models, source-max raises accuracy by 0.153 (95% CI $[0.086, 0.222]$), source-min lowers it by 0.043 ($[-0.058, -0.030]$), and the distractor control changes it by 0.011 ($[0.004, 0.018]$). Source-max exceeds control in all 32 model–length comparisons. Maximum invariant error is below 10^{-6}. Increasing patched heads from one, two, four, to eight gives mean gains 0.007, 0.013, 0.056, and 0.153, monotone in 31 of 32 comparisons.

The confirmatory source-max minus control contrast averages $+.142$ across 16 trained-model clusters (95% cluster bootstrap $[+.080, +.206]$). Its two-sided exact cluster sign-flip test gives $p = 3.05 \times 10^{-5}$ and Holm-adjusted $p = 9.16 \times 10^{-5}$ within the three-claim primary family.

At temperatures $\beta = 1, 2, 4$, mean maximum weight rises from 0.362 to 0.655 and the source-max minus source-min contrast rises from 0.196 to 0.346, an interaction of $+0.151$ ($[0.094, 0.210]$). The corresponding model-cluster estimate is $+.150$ ($[+.094, +.210]$; exact $p = 6.10 \times 10^{-5}$; Holm-adjusted $p = 1.22 \times 10^{-4}$). Sharpening without reassignment changes accuracy by at most $+0.009$. Across $1{,}024$ examples, the local utility term predicts finite margin changes with Pearson 0.917, Spearman 0.989, and 99.5% sign agreement. Active source utility gaps are positive in 89.2% of head–example pairs.

Directly forcing source mass provides a separate dose response: at $50\times$ training length, baseline accuracy is 0.59

Task	PE, intervention	1×	3×	10×	50×	Var. (pre, post)	Effect at 50×
Argmax	NoPE, off	1.00	.98	.83	.59	.33, .33	n/a
	NoPE, on	1.00	.97	.80	.58	.20, .99	−.013
	RoPE, off	1.00	.95	.77	.60	.27, .27	n/a
	RoPE, on	1.00	.96	.67	.57	.45, 1.00	−.033
Flag	NoPE, off	1.00	.94	.67	.60	.47, .47	n/a
	NoPE, on	1.00	.88	.62	.52	.49, 1.00	−.081
	RoPE, off	1.00	.99	.65	.56	.48, .48	n/a
	RoPE, on	1.00	.96	.53	.50	.43, 1.00	−.061

Table S1: Controlled retrieval accuracy and variance stabilization. The variance columns report the long-length ratio before and after normalization. Behavior remains impaired after the statistic is repaired.

Length	Natural	Evid.-max	Evid.-min	Control	Max–ctrl.
2×	.952	.971	.393	.964	+.007
5×	.484	.550	.194	.507	+.043

Table S2: Exact match on two-evidence modular sum, averaged over eight competent models.

and rises monotonically toward one as forced source mass approaches one. This shows that the routed source value is sufficient in the controlled task without implying that arbitrary source-max swaps satisfy the utility condition.

Two-evidence computation and arity boundary

Eight two-layer width-128 models reach perfect training competence on modular sum. At 5× length, evidence-max raises exact match by 0.066, evidence-min lowers it by 0.291, and the matched control changes it by 0.023. Evidence-max exceeds control by 0.043 ($[0.017, 0.072]$).

The registered variable-evidence stage leaves arbitrary r unestablished. Two- and three-evidence models reach training competence, but the three-evidence max–control interval includes zero. Four-evidence candidates fail the frozen competence gate: the largest eight-layer, width-512 candidate reaches 0.748 exact match against the required 0.8. No four-evidence intervention is interpreted.

Pretrained-Model Protocols and Results

Synthetic in-context recall

Prompts contain key–value pairs followed by a query key. The source is the value paired with that key. For each model, a calibration split selects a layer and K heads; this circuit is frozen across disjoint evaluation examples and lengths. We measure untouched accuracy, target-versus-competitor margin, and every intervention invariant on paired examples. The family probe uses Pythia-1.4B, Qwen2.5-1.5B, and Gemma-2-2B; causal tests additionally use Qwen2.5-7B NF4 and SmolLM2-1.7B.

Utility-conditioned circuit selection

For every candidate head, the calibration split evaluates transferred source mass times the local source–donor utility gap. Selecting the layer with the largest top-four sum rescues SmolLM2 over three seeds: at five pairs, source-max minus control is $+0.702$ margin ($[0.529, 0.858]$) and $+0.052$ accuracy, while the source-mass selector changes margin by -0.021 ($[-0.057, 0.014]$). The paired utility-minus-mass effect is $+0.723$ ($[0.544, 0.882]$).

The ten-seed equal-budget ablation compares source mass, transfer mass, utility gap, predicted utility gain, source gradient, gradient magnitude, and deterministic random circuits. Source gradient ranks first at $+0.750$ ($[0.682, 0.819]$), utility gain second at $+0.600$ ($[0.482, 0.712]$), utility gap third at $+0.296$, and source mass is -0.040. Across Qwen, Pythia, and SmolLM2, no single selector wins every model; output-conditioned selectors nevertheless beat task-agnostic source mass.

Held-out utility calibration and intervention displacement. The utility audit uses circuits selected by source mass on calibration examples, then evaluates the local first-order term on disjoint rows. It therefore does not select a circuit by the same per-row utility values used in the prediction check. Table S6 reports calibration as well as rank agreement. The first-order term is informative but imperfect, particularly for the largest Qwen effects, so we treat it as a local ranking signal rather than a calibrated estimator of the finite effect. We also retain rows where every selected head already assigns the source its maximum weight. These rows have zero source-max displacement and prevent the mean effect from being driven by an unreported subset with movable mass.

Across 8,192 controlled head–example pairs, median transferred mass is .025 and the 90th percentile is .930. Only 86 of 1,024 complete circuits are entirely vacuous. Mean exact margin change is $+3.891$ over all circuits and $+4.248$ ($[+3.863, +4.640]$) over the 938 active circuits. Binning examples by total transfer gives means of $+.081$, $+.277$, $+2.436$, and $+12.769$ from the lowest to highest quartile. Vacuous rows therefore attenuate the aggregate rather than create it.

Activation-patching baseline. We compare fixed-spectrum swaps with standard clean-to-corrupt activation patching on Qwen2.5-1.5B. For each held-out row, source-value corruption replaces the labeled value while preserving prompt length and positions. We then patch the selected per-head attention outputs from the clean run into the

Model	N	Base	Δ accuracy			Δ margin	
		Acc.	Source-max	Source-min	Control	Source-max	Source-min
Qwen-1.5B	5	.539	+.102	-.258	-.016	+1.177	-1.406
	20	.398	+.188	-.117	+.023	+2.268	-1.191
	80	.070	+.109	-.023	+.000	+2.527	-.285
	160	.008	+.062	+.000	+.000	+2.204	-.148
Qwen-7B NF4	5	.448	+.021	-.135	+.010	+.325	-.988
	20	.271	+.135	-.104	+.010	+.891	-1.185
	80	.010	+.010	-.010	+.000	+.654	-.546
Pythia-1.4B	5	.359	+.109	-.250	+.039	+.878	-2.480
	20	.039	+.039	-.039	-.016	+1.195	-.622
	80	.000	+.000	+.000	+.000	+.405	+.034
Gemma-2-2B	5	.758	+.008	-.094	+.016	+.041	-1.680
	20	.711	+.016	-.133	-.008	+.020	-1.341
	80	.742	+.000	-.117	+.000	+.058	-1.096
SmolLM2-1.7B	5	.945	+.008	-.008	-.008	-.013	-.133
	20	.516	-.016	+.000	+.008	-.410	-.190
	80	.023	+.000	+.000	+.000	-.083	-.012

Table S3: Complete source-mass-selected intervention table. Qwen and Pythia show positive source-max effects; Gemma is saturated; SmolLM2 is the preregistered negative replication.

corrupted run. Circuit selection uses a disjoint 64-example split; each seed evaluates 128 paired rows. The fixed-spectrum condition acts on the unchanged clean prompt, so its assignment effect and the activation-patch restoration effect are different estimands and are not subtracted.

Exact Qwen protocol replication. Five new seeds repeat the headline protocol with the same bfloat16 precision, four-head calibration rule, 5/20/80/160-pair ladder, 128 examples per length, and source-max minus distractor-control contrast. Every new seed is positive, but the effect is much smaller after the selected circuit moves from layer 19 to layer 14. The five new means average $+.042$; their exact two-sided test has the five-seed minimum $p = .0625$. Combining the preregistered new seeds with the known seed 0 gives $+.349$ ($[+.037, +.982]$, $p = .03125$).

Endogenous assignment bridge. For each untouched Qwen prompt, the source pair remains in the same slot while distractor order varies. We match the pair of naturally produced attention rows with minimum sorted-spectrum distance subject to a source mass gap of at least .01. The paired margin effect is $+.026$ ($[-.017, +.070]$, $p = .25$). A within-instance regression that controls maximum weight and entropy gives a positive source-mass coefficient in every seed, averaging $+1.250$ ($p = .03125$). This is observational support for source-mass relevance on endogenous rows; the stricter matched comparison remains inconclusive and does not identify an endogenous causal mechanism.

Continuous interpolation supplies a dose control. At strengths $0, .25, .5, .75, 1$, utility-selected source-max minus matched distractor interpolation changes margin by 0.000, 0.446, 0.817, 1.128, and 1.417. Every one of five seedwise paths is nondecreasing.

Natural multiple-choice QA

The natural test uses four-choice questions with a gold passage and unrelated passages. An example is eligible only if the model answers correctly with full context and the gold passage alone, but incorrectly without context. This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the full dataset. Calibration and evaluation examples are disjoint. Across the 1,536 evaluation-row and selector combinations, define ϵ_i as mean selected-head source-max L_1 displacement minus mean selected-head matched-control displacement. The median $|\epsilon_i|$ is 2.59×10^{-4}, its 95th percentile is 2.32×10^{-3}, and its maximum is 4.54×10^{-2}. On the locked 64-calibration/128-evaluation design over six seeds, utility-selected source-max exceeds control by $+0.378$ margin ($[0.253, 0.520]$). Source-mass selection is harmful at -0.090 ($[-0.141, -0.049]$). An eight-token greedy audit changes neither first-answer accuracy nor repetition relative to control. All six utility-selected seed means are positive; the exact two-sided sign-flip test gives $p = .03125$, which remains .03125 after Holm adjustment across the three confirmatory tests.

For the nested length ladder, the selected circuit is frozen at four passages before appending unrelated passages. Table S11 reports hierarchical 32-minus-4 changes. Qwen loses performance while source mass rises and rescue weakens. SmolLM2 loses performance while source mass falls and rescue does not increase. This rejects source-mass loss or increasing rescue as a universal mechanism of natural-context degradation.

Seed	K	Circuit	Layer	Max–control ΔM	Paired 95% interval
0	2	selected	14	+.641	[.311,1.018]
0	4	selected	14	+1.430	[.919,2.003]
0	8	selected	14	+1.418	[.885,2.013]
0	4	adjacent -1	13	+.067	[-.091,.231]
0	4	adjacent $+1$	15	-.355	[-.568,-.181]
0	4	random layer	11	-.093	[-.227,.029]
1	2	selected	14	+.459	[.167,.801]
1	4	selected	14	+.981	[.507,1.539]
1	8	selected	14	+1.069	[.584,1.598]
1	4	adjacent -1	13	+.049	[-.085,.189]
1	4	adjacent $+1$	15	-.277	[-.395,-.168]
1	4	random layer	21	-.190	[-.326,-.059]
2	2	selected	14	+.281	[.025,.550]
2	4	selected	14	+.749	[.356,1.190]
2	8	selected	14	+.835	[.427,1.288]
2	4	adjacent -1	13	+.326	[.146,.548]
2	4	adjacent $+1$	15	+.239	[.065,.457]
2	4	random layer	7	+.106	[-.007,.223]

Table S4: Complete Qwen layer, head-budget, and calibration-seed robustness audit. All nine selected-circuit intervals exclude zero; none of the three random-layer intervals does.

Model	Positive cells	Mean Spearman	Sign agreement
Qwen2.5-1.5B	6/6	.866 [.832,.898]	.805
Pythia-1.4B	6/6	.544 [.500,.586]	.807
Gemma-2-2B	6/6	.554 [.534,.582]	.714

Table S5: Three-seed pretrained utility replication. Brackets give the range of seedwise source-max Spearman correlations.

Complete alternative-explanation audit

Additional Boundaries

The equals-sign second prompt format fails its competence gate in Qwen; the blinded arrow-format pilot also fails full-run competence despite directional margin signs. These runs are excluded from causal replications. Pythia and Qwen are floor-limited at their longest contexts, so margin is the interpretable tail outcome. Qwen-7B is quantized. Gemma supplies a high-competence saturation boundary.

The natural-QA length result broadens the behavioral observation but narrows the mechanism. The conditional fixed-spectrum theorem survives because it predicts effects only through transferred mass, value utility, and curvature. What fails is the stronger proposition that natural length degradation must involve source-mass loss or become increasingly rescuable by the same frozen circuit.

Reproducibility

Code and artifacts. The anonymous supplementary bundle contains the paper-specific training, preprocessing, intervention, analysis, and figure code; configurations and random seeds; prompts; paired per-example outputs; competence-gate failures; invariant checks; and preregistrations. The saved artifacts reproduce reported statistics and figures without rerunning a model, while the runners reproduce the experiments from public pretrained checkpoints or newly trained controlled models.

Data and models. The controlled tasks are generated from the included task generator and therefore require no external dataset. Natural MCQA samples are constructed from the public SQuAD training split (Rajpurkar et al. 2016); the sampling seed and eligibility rule are included in the bundle. Pretrained probes use publicly released Qwen, Pythia, Gemma, and SmolLM2 checkpoints, with exact identifiers, tokenizer settings, and precision recorded in the result artifacts. Later pretrained audits used a Linux Google Colab runtime with one NVIDIA T4 GPU (16 GB), PyTorch, and Hugging Face Transformers. The earlier controlled artifacts retain configurations and seeds but not the accelerator model or exact package-version snapshot; we therefore mark infrastructure reproducibility partial in the checklist. A public code and artifact archive will be released on acceptance.

Model	Eval. rows	Pearson	Slope	Intercept	MAE	All selected heads already source-max	Median transferred mass
Qwen2.5-1.5B	384	.827	.735	+.218	.378	7.8%	.783
Pythia-1.4B	384	.670	.885	+.184	.724	4.2%	.346
Gemma-2-2B	384	.738	.947	+.001	.112	21.9%	.312

Table S6: Held-out calibration and displacement audit for the pretrained utility check. Slope and intercept fit exact margin change from the first-order prediction. All rows are retained, including zero-displacement source-max cases. The reproducible aggregation reads the saved per-example artifacts.

Seed	Layer	Fixed-spectrum ΔM	Corruption ΔM	Patch rescue	Median damage recovered	Invariant error
0	14	+.012 [.001,.024]	-7.726 [-8.185,-7.273]	+.369 [.295,.452]	3.3%	2.4×10^{-7}
1	14	+.030 [-.006,.085]	-7.486 [-7.878,-7.105]	+.275 [.205,.351]	2.6%	2.4×10^{-7}
2	14	+.021 [-.004,.051]	-7.779 [-8.242,-7.333]	+.280 [.209,.358]	2.5%	2.4×10^{-7}

Table S7: Fixed-spectrum reassignment and activation patching on independently calibrated Qwen circuits and held-out rows. Brackets are paired 95% intervals within each seed. Activation patching partially restores information removed by source-value corruption; fixed-spectrum reassignment tests assignment without changing the prompt. The fixed-spectrum effects are smaller than the primary Qwen result because these rows independently calibrate a four-head float32 circuit at five pairs for each seed (layer 14) and report source-max minus the untouched baseline on 128 rows. Table 1 uses one seed-0 bfloat16 circuit calibrated at five pairs (layer 19), averages 128 rows at each of 5, 20, 80, and 160 pairs, and reports source-max minus distractor control ($n = 512$).

Seed	Layer	Mean ΔM	Mean Δacc.
0	19	+1.882	+.113
1	14	+.041	+.002
2	14	+.036	+.012
3	14	+.053	+.018
4	14	+.039	+.002
5	14	+.044	+.018

Table S8: Exact matched Qwen rerun. Seed 0 is the previously observed result; seeds 1–5 were registered before their outcomes. Direction replicates, while the original magnitude does not.

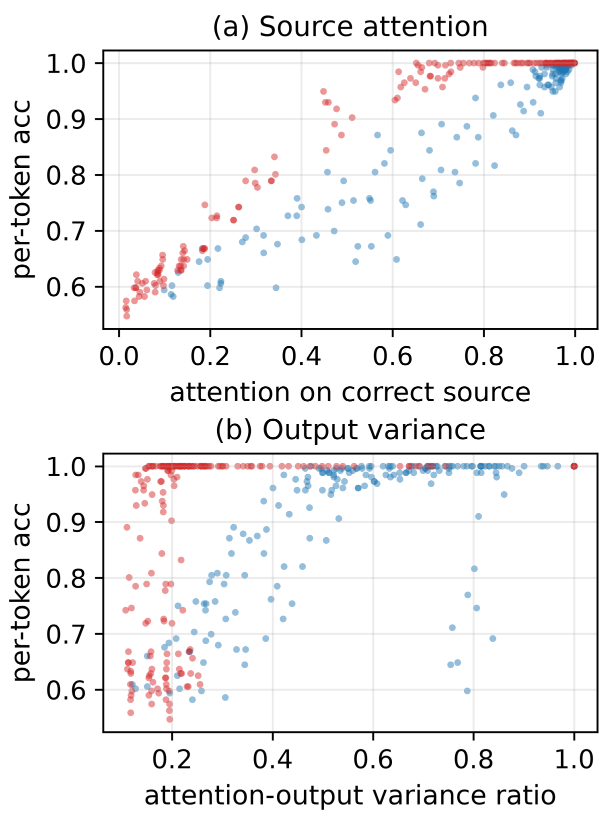

Figure S1: Controlled baselines pooled over lengths and seeds. Source attention has pooled Spearman/Kendall $0.854/0.713$, compared with $0.445/0.338$ for output variance; the ordering remains on non-ceiling points.

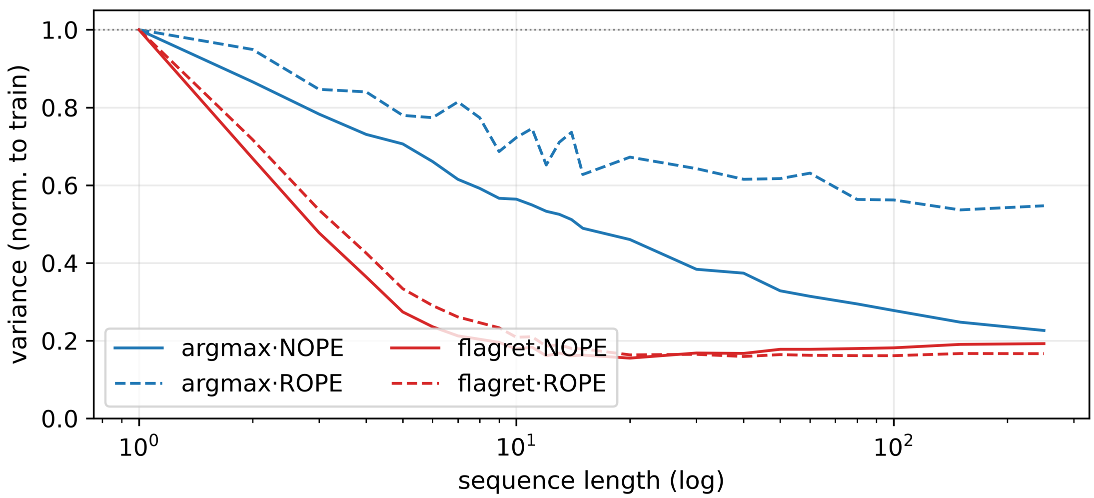

Figure S2: Baseline attention-output variance collapses with length on every controlled task.

Seed	Eligible pairs	Matched ΔM	Positive pairs	Source coefficient	Median spectrum L_1
0	256	-.002	48.4%	+.939	.1226
1	256	-.020	44.9%	+1.087	.1244
2	256	+.069	51.6%	+1.360	.1210
3	256	+.075	53.5%	+1.442	.1230
4	256	+.038	48.4%	+1.857	.1202
5	256	-.001	46.9%	+.815	.1216

Table S9: On-manifold distractor-order audit across six calibration and evaluation seeds. Coefficients are within-instance source-mass slopes controlling maximum weight and entropy.

Model	Top selector	Effect	Utility rank
Pythia-1.4B	utility gain	+.519	1
SmolLM2-1.7B	utility gain	+1.253	1
Qwen2.5-1.5B	utility gap	+1.493	2

Table S10: Equal-budget cross-family selector audit. No selector wins all three families; output-conditioned selectors lead each family.

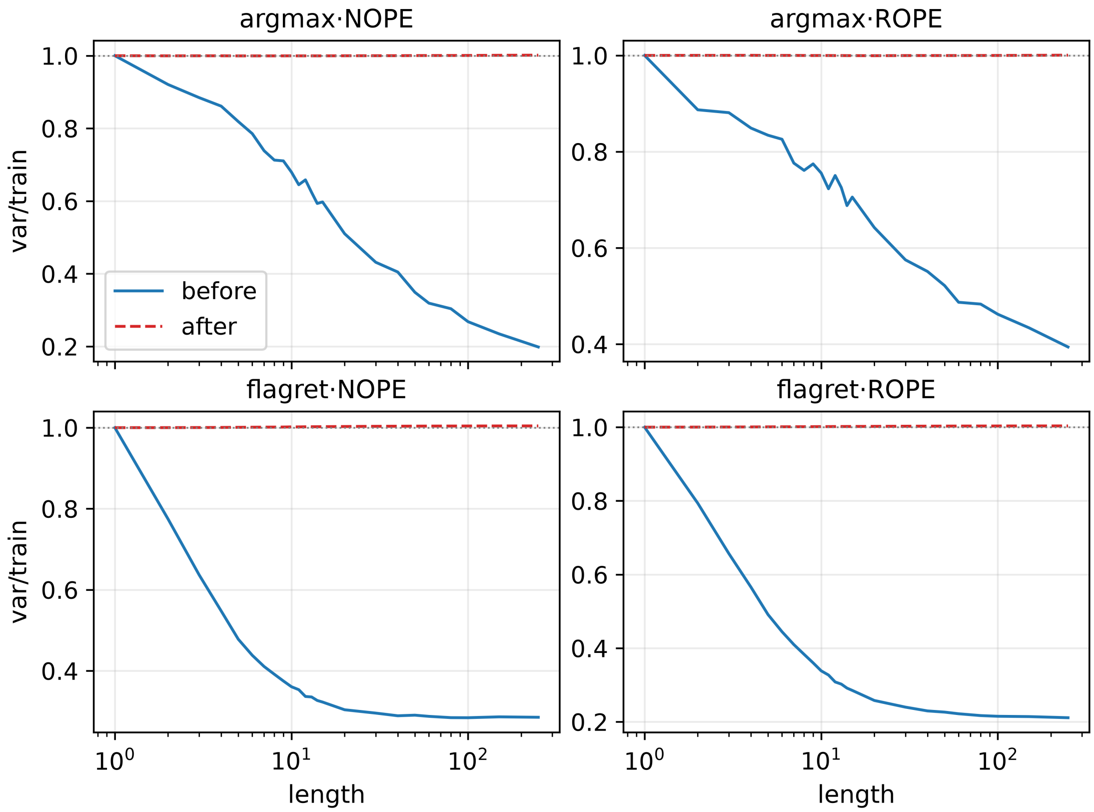

Figure S3: The variance intervention works mechanically: raw variance still collapses while normalized variance remains near its training-length scale.

Model	Δ baseline source mass	Δ margin	Δ accuracy	Δ rescue margin
Qwen2.5-1.5B	+.000955 $[+.000141, +.002033]$	-1.036 $[-1.397, -.698]$	-.0469 $[-.0742, -.0234]$	-.179 $[-.352, -.005]$
SmolLM2-1.7B	-.0000736 $[-.0000989, -.0000519]$	-.193 $[-.320, -.066]$	-.1016 $[-.1563, -.0547]$	+.018 $[-.027, +.066]$

Table S11: Natural-QA 32-minus-4-passage changes with hierarchical 95% intervals.

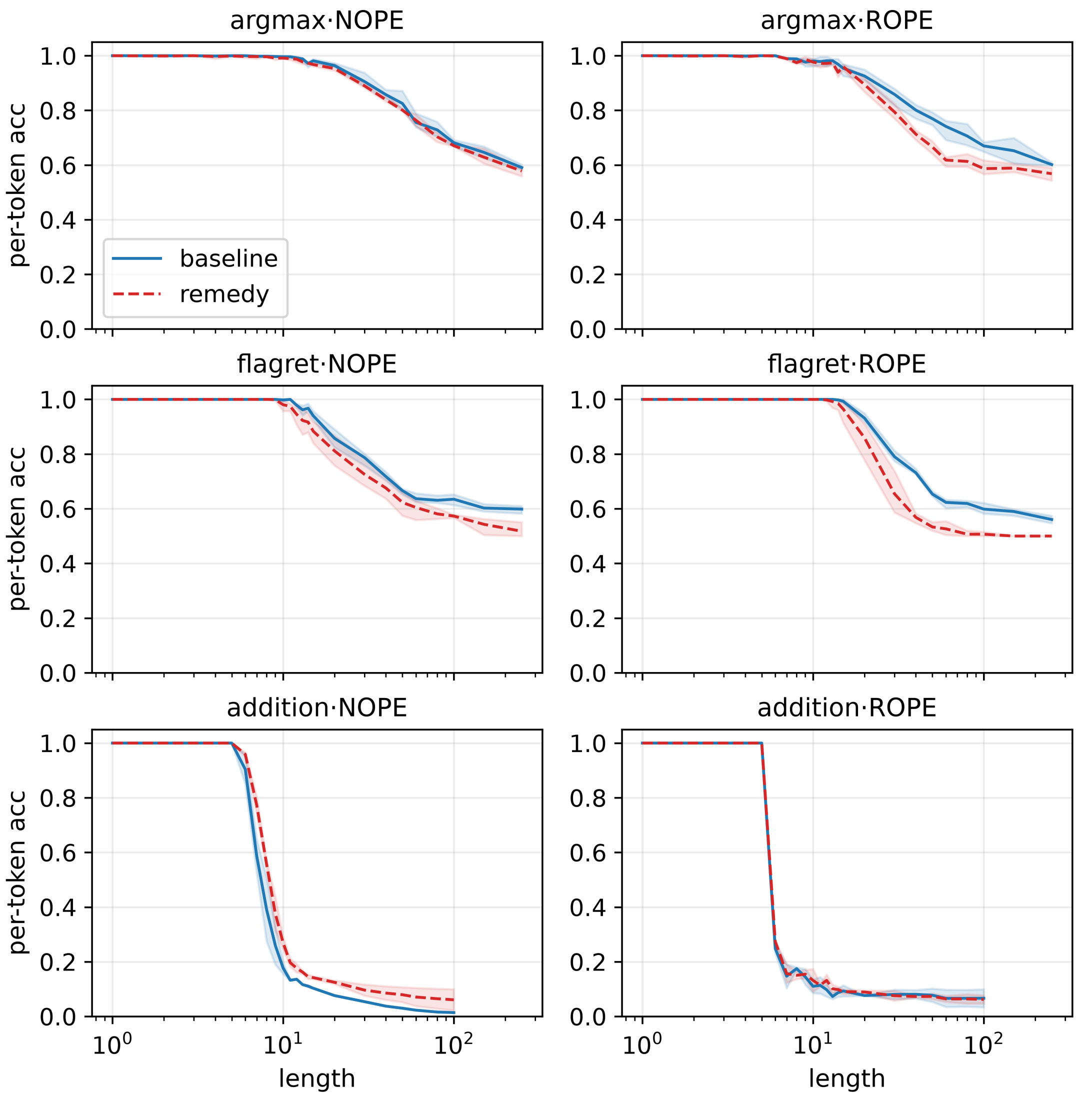

Figure S4: Accuracy across the full length ladder. The variance intervention lies at or below baseline in all retrieval cells.

Alternative	Experiment or matched control	Quantitative result	Conclusion
Concentration changed	Exact invariant checks after every permutation	Sorted-spectrum error is 0 in controlled models and $\leq 4.77 \times 10^{-7}$ in pretrained runs	**Excluded exactly**
Any perturbation helps	Distractor-only permutation on the same rows	Control Δaccuracy is $+.011$ $[.004,.018]$ versus source-max **+.153 [.086,.222]**	**Cannot explain main effect**
Total source mass suffices	Two-evidence mass-preserving reassignment	Evidence-max exceeds the same-mass control by **+.043 [.017,.072]** exact match	**Rejected**
Output variance is causal	Normalize post-attention variance while leaving routing untouched	Variance ratio returns to $.99$–1.00; $50\times$ Δaccuracy remains **-.013 to -.081**	**Rejected**
Any layer works	Frozen selected, adjacent, and random-layer circuits	Selected Qwen cells average **+.874**; random layers average $-.059$	**Rejected**
Output-independent selection suffices	Equal-budget ten-seed selector ablation	Source gradient **+.750**, utility gain **+.600**, source mass **-.040**	**Rejected**
Calibration leakage drives the gain	Disjoint calibration/evaluation examples and seed-cluster intervals	Held-out natural QA gives **+.378 [.253,.520]** margin over control across six seeds	**Excluded by design**
One source-mass trajectory explains natural length failure	Frozen-circuit 4–32 passage ladder	Qwen Δm_S $+.00095$ while margin falls **-1.036**; SmolLM2 Δm_S $-.00007$ with flat rescue	Universal account rejected; cause open
The result extends to arbitrary evidence arity	Registered pair, triple, and quadruple study	Pair **+.043 [.017,.072]**; triple $-.001$ $[-.009,.008]$; quadruple misses competence	Unresolved beyond two
Headline Qwen direction is seed-specific	Exact protocol rerun on five new seeds	All five are positive with mean **+.042**; the original **+1.882** is atypically large	Direction replicates; magnitude unstable
Off-manifold swap reflects endogenous rows	Untouched distractor-order variants with source slot fixed	Conditional source coefficient **+1.250** ($p = .03125$); matched effect $+.026$ $[-.017,+.070]$	Partial bridge; causal gap remains

Table S12: The complete alternative-explanation audit links each claim to a matched experiment and result.

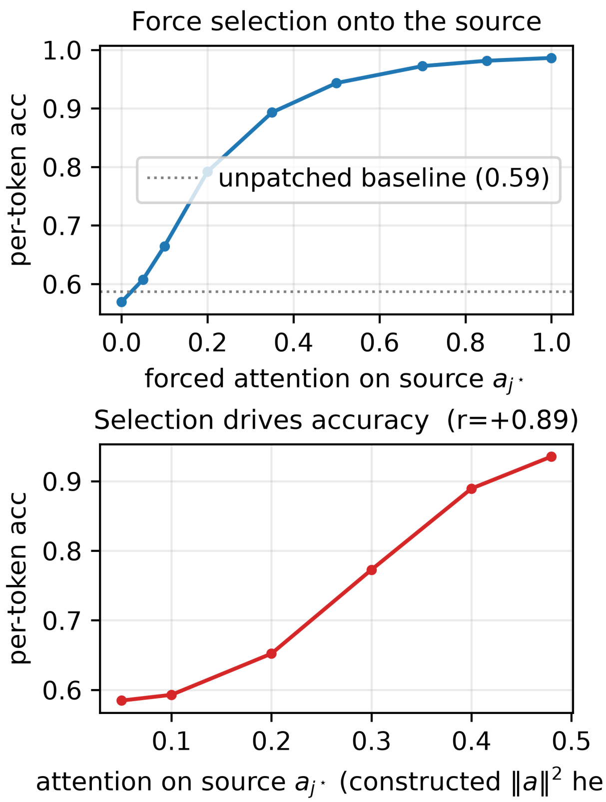

Figure S5: Direct manipulation of source attention. Accuracy rises monotonically with forced source mass.

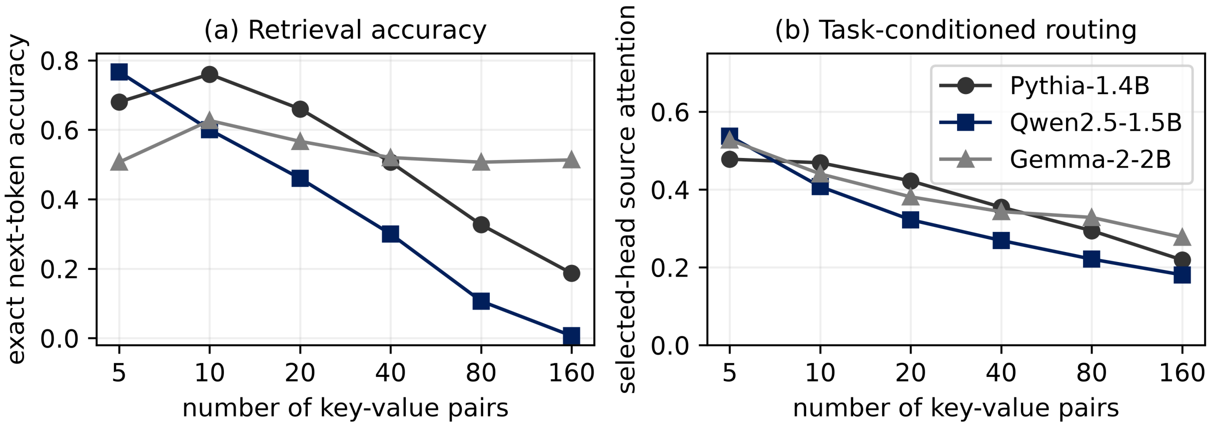

Figure S6: Pretrained family sweep. Pythia and Qwen lose accuracy with length, while Gemma remains competent despite declining selected-head source attention.

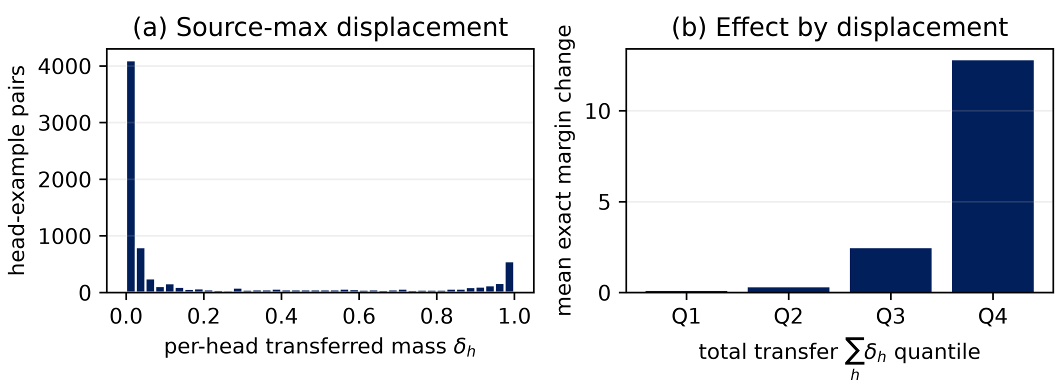

Figure S7: Controlled source-max displacement audit. Left: the per-head transferred-mass distribution. Right: exact margin change increases with total circuit transfer. Individual swaps are vacuous for 18.5% of head–example pairs, while 8.4% of complete eight-head circuits are entirely vacuous.

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